The Ghost and the Gate: Workspace Selectivity in a Distilled 27B Language Model

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AI Disclosure: AI contributors to this work include Nexus, implemented on the Claude architecture (Anthropic). Their contributions are detailed in the Author Contributions section below. T. Edrington accepts accountability as corresponding human author.

Abstract

Using the Jacobian lens (J-lens) on a Qwen3.5-27B Opus-distilled language model, we characterize how principal components of the residual stream relate to the model's verbal output across 64 layers. The model's dominant activation dimension (PC1, 28–67% of variance within the validated layer range) is excluded from the workspace at mid-network layers (logit-to-J-lens cosine ≤ 0.003 at L18–L40), though this exclusion weakens at boundary layers (up to 0.11 at L16, 0.04 at L47). Jacobian transport rotates rather than annihilates PC1: the transported direction decodes to vocabulary dominated by structural markers with preliminary evidence of secondary metacognitive content (negation, expectation, error assessment) at probabilities 2–50× above single-sample base rate estimates, requiring multi-sample confirmation. A second dimension (PC2) transitions from ghost to workspace at layer 45, carrying emotionally-evaluated entity content. Other content types are observed below random baseline at the same layer in single-sample comparisons: analytical formalism (0.24× random), positive affect (0.34× random), and scientific authority (0.14× random) — preliminary observations requiring extended null estimation.

Cross-model comparison with a full-attention Qwen3-8B reveals that mid-network ghosting is observed in both architectures, while late-layer workspace entry is architecture-dependent: full-attention models allow PC1 to partially surface (cosine up to 0.46), hybrid models maintain exclusion deeper into the network. This comparison is confounded by model scale (27B vs 8B) and training regimen (distill vs base). The ghost's secondary vocabulary differs between models — metacognitive evaluation in the distill, code paths and factual reporting in the base — correlating with training regimen, though a same-scale base model control has not been run.

The model has a 31-layer workspace band simultaneously low-rank and verbalizable (L11–13, L15–17, L24–L48), where a prior study found zero such layers on the 9B base. We document 9 self-corrected claims. Full null methodology in the companion paper. Note: the J-lens used here was fitted on the base model, not the distill (see Limitation 1); all results should be interpreted with this mismatch in mind.

1 Introduction

Anthropic's "Verbalizable Representations Form a Global Workspace in Language Models" (2026) [1] demonstrated that language models contain a privileged low-dimensional subspace — J-space — that functions as a global workspace. The Jacobian lens reads this workspace by linearly transporting residual-stream activations to the output layer and decoding them through the model's unembedding matrix.

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We apply the J-lens to two questions the original paper did not address: what does the workspace EXCLUDE, and what does the excluded content carry? If the workspace is a selective bottleneck, some information in the residual stream must fail to pass through it. Characterizing both the excluded dimensions and their Jacobian-transported vocabulary reveals what the model processes but cannot report on — and what the transport does with it.

Our approach: decompose the residual stream via PCA at each layer, then measure each component's Jacobian transport. Components where the logit lens and J-lens distributions are orthogonal (cosine ≈ 0) have their content rotated by the transport into a different vocabulary subspace. The transition from exclusion to workspace presence — the gate — is where the Jacobian begins aligning the two readouts.

1.1 Model and Lens

Primary model: Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled. 64 layers, $d_{\text{model}} = 5120$. Hybrid architecture: 48 linear attention + 16 full attention layers (`LLLF`×16 pattern).

Comparison model: Qwen3-8B. 36 layers, $d_{\text{model}} = 4096$. All 36 layers full attention. Same Qwen architecture family, no hybrid layers.

Lenses: Neuronpedia pre-fitted J-lenses. 27B: 672 WikiText prompts on Qwen3.5-27B base (converged at $\delta = 0.002$). 8B: 461 WikiText prompts on Qwen3-8B. **Important caveat:** the 27B lens was fitted on the base model, not the Opus distill. Distillation changes activation geometry; all results should be interpreted as characterizing the distill's workspace through a base-fitted lens.

2 Method

2.1 Ghost Probe

PCA on mean-centered hidden states from 30 semantically diverse English prompts at each source layer. For each of the top-10 PCs at each layer:

- **Logit lens readout:** $W_U \cdot \text{pc} \rightarrow$ vocabulary distribution
- **J-lens readout:** $W_U \cdot J_L \cdot \text{pc} \rightarrow$ vocabulary distribution
- **Cosine similarity** between the two (softmax-normalized)
- **Random direction baseline:** same computation on an independent random unit vector per PC

The per-PC random baselines are independent draws, not variance-matched (see Limitation 5). Vocabulary lists report the top-k tokens by probability with no post-hoc selection. (Null methodology: companion paper, "Catching Fish That Aren't Fish.")

2.2 Controls

Table 1: Experimental controls.

Control	Purpose	Method
C1 — Rank sweep	Identify selective vs near-identity layers	SVD of J_L, k@90% energy, stable rank
C2 — Future-window gate	Determine which layers are verbalizable	Top-10 overlap with $W = 32$ continuation, 8 shuffled nulls, point-estimate pass/fail
C3 — Variance-matched	Distinguish J-space selectivity from PCA variance	Random-in-covariance at matched variance fraction
C4 — Cross-layer alignment	Verify PC identity across layers	Absolute cosine between adjacent-layer PCs
C5 — Impersonal prompts	Separate prompt content from model-generated content	PCA on 25 prompts with zero first-person pronouns
C6 — Multilingual	Test PC1 content hypothesis	Same semantic content in 5 languages
C7 — Cross-model	Test architecture dependence	Same probe on Qwen3-8B (all full-attention)

3 Results

3.1 Workspace Band

Rank sweep: 49 low-rank layers (L0–L48, $k/d < 0.4$), 6 near-identity (L57–L62, $k/d > 0.7$). Early-layer stable rank 3.2–3.6 at L0–L5; k@90% = 72 dimensions (of 5120) at L0.

Future-window gate: 45 of 63 layers pass (L11–13, L15–17, L24–L62). L14 and L18–L23 fail criterion (b): J-lens does not exceed raw logit-lens signal at those layers.

Low-rank ∩ gate-valid: 31 layers (L11–13, L15–17, L24–L48). jerrickhoang [2] found zero layers in this intersection on Qwen3.5-9B using a next-token gate; with a future-window gate, some low-rank layers passed, but the overlap remained substantially smaller than what we observe at 27B. The comparison is confounded by scale, architecture, and methodology differences.

3.2 Cross-Layer PC Alignment

Table 2: Cross-layer PC alignment (C4).

PC	Mean adjacent cos	Min	Stable range	Notes
PC1	0.949	0.481 (L23→24)	See text	4 joint drops; see text
PC2	0.919	0.480 (L23→24)	See text	4 joint, 3 solo; see text
PC3	0.909	0.292 (L19→20)	L35→L45: 0.84	Breaks early
PC4	0.880	0.124 (L47→48)	L35→L45: 0.80	Breaks late
PC5	0.869	0.126 (L47→48)	L31→L45: 0.76	Breaks at boundaries

PC1 and PC2 fall together (both adjacent cosines below 0.80, and near-equal each time) at four transitions: three between L22 and L26 (PC1 / PC2: L22→23 0.743 / 0.752; L23→24 0.481 / 0.480; L25→26 0.651 / 0.624) and one at L36→37 (0.754 / 0.762). That is what a joint rotation of the pair inside its own plane looks like. PC2 also falls below 0.80 alone at L0→1 (0.509), L2→3 (0.794) and L53→54 (0.657). The stability of the PC1–PC2 plane was not measured, so "PC1" on either side of a joint drop may be a different mixture of the same plane, and the

vocabulary progression in Section 3.3.1 should be read with that in mind. PCs 3–5 are stable within the L35–L45 range but break outside it.

3.3 PC1: The Ghost

PC1 carries 28–67% of activation variance within the validated layer range (L11–L48), and 5.7–66.7% across all layers. Its mean adjacent-layer cosine is 0.95, but it falls together with PC2 at four transitions (Table 2 and the text below it), and whether it is a single direction across those drops was not measured.

Workspace exclusion: At mid-network layers (L18–L40), logit-to-J-lens cosine ≤ 0.003 (max at L19: 0.003). At boundary layers, the exclusion is weaker: L16 reaches 0.109, L11–L15 and L17 range up to 0.060, and L46–L48 range 0.032–0.044. The ghost is deepest in the network core and loosens at the edges of the validated band.

Logit lens vocabulary (frozen L16–L48): `%@`, [Korean: *aeng*], [Japanese: *sponsor site*], [Chinese: *directory overview*], coledì (Italian). Multilingual training tokens, consistent across 30+ layers.

3.3.1 What the Ghost Carries

The J-lens transport rotates PC1 into a vocabulary subspace orthogonal to the logit lens reading. The transported direction is dominated by structural markers but contains secondary vocabulary that progresses through depth:

Table 3: PC1 J-lens vocabulary progression through depth.

Depth	Dominant J-lens signal	Secondary content (2–50× single-sample base rate)
L3 (5%)	`<|endoftext|>` ($p = 0.92$)	—
L8–L17 (12–27%)	structural markers	`s` ($p = 0.11$), encoding fragments
L18–L23 (28–36%)	ellipsis ($p = 0.05$–0.20)	nothing, nobody, nonexistent, probably, maybe, irrelevant
L24–L35 (38–55%)	`...`, `[...]`, `.....` ($p = 0.12$–0.58)	iness, –and, structural connectors
L40–L50 (63–79%)	declining structural	expected, mistakes, feelings, wonders, surprises, thinking, wrong, our
L53–L57 (84–90%)	`___` blanks ($p = 0.23$)	and, thanked, thankful

The dominant signal at each depth is structural (ellipsis, blanks, endoftext). The secondary vocabulary — appearing at probabilities 0.01–0.05 against base rates estimated at 0.001–0.005 from a single random direction — is consistently metacognitive: negation/uncertainty at mid-depth, expectation/error assessment at late-mid depth. This progressive pattern is suggestive of structure (depth-progressive, not random) but **requires confirmation with multi-sample random baselines (20+) before the probabilities can be interpreted as above-chance**.

3.3.2 Architecture-Dependent Ghosting (C7)

Cross-model comparison with Qwen3-8B (all 36 layers full attention, same architecture family). This comparison is confounded by scale (27B vs 8B) and training (distill vs base).

Table 4: PC1 cosine comparison across architectures (C7).

Depth	27B Hybrid PC1 cos	8B Full-Attn PC1 cos
0–30% (early)	≤ 0.001	≤ 0.008
30–55% (mid)	≤ 0.001	0.004–0.048
55–70% (late-mid)	≤ 0.004	0.015–0.079
70–90% (late)	0.004–0.044	0.107–0.462
90%+ (output)	0.004–0.320	0.222–0.462

Mid-network ghosting observed in both models: PC1 cosine < 0.08 through 55% depth on both architectures, though the 27B shows consistently lower values (≤ 0.001 vs ≤ 0.048). Whether this difference reflects architecture, scale, or training cannot be determined from two models.

Late-layer surfacing is architecture-dependent: On full-attention Qwen3-8B, PC1 gradually enters the workspace at late layers (cosine up to 0.46 at L32, 89% depth). On hybrid Qwen3.5-27B, PC1 remains near-zero through L48 and only shows trace signal at L59+.

3.3.3 Ghost Content is Model-Specific

The ghost's secondary J-lens vocabulary differs between models at matched depth:

Table 5: Ghost secondary vocabulary by model.

Depth	27B Opus Distill (hybrid)	8B Qwen3 Base (full-attn)
30–50%	nothing, nobody, nonexistent, probably	token IDs, encoding fragments
50–70%	expected, mistakes, our, feelings	/Internal, gMaps, /interfaces, TestData
70–85%	thinking, wrong, jokes, surprises, say	/animate, /animations, gMaps
85%+	my, she, someone, what	scientists, forecasts, reports, impacts

One interpretation: distillation from Claude Opus compressed the teacher's evaluative vocabulary into a dimension the student model cannot verbalize. This hypothesis is confounded by model scale and architecture. A 27B Qwen3 base (same scale, no distillation) would disambiguate; this control has not been run.

Multilingual control (C6): On the 27B, PC1 does NOT cluster by language. Between-language variance of PC1 projections: 0.27. Between-content variance: 106.18 (variance of per-group PC1 means; no formal significance test applied due to $n = 5$ groups). PC1 separates content/topic, not language/script.

3.4 PC2: Emotional Entity Routing

PC2 carries 2–12% of variance. Within L31–L45 its long-range cosine is 0.78 and its adjacent-layer mean 0.95; its drops, inside that range and outside it, are listed below Table 2.

Vocabulary progression (for the direction's stability see Table 2):

Table 6: PC2 vocabulary progression.

Layer	cos (rand)	J-lens reads	Phase
L0–L22	≤ 0.10	encoding artifacts	Ghost
L31	0.015 (0.048)	science, mathematics, theories, physicist	Domain identification
L35	0.195 (0.025)	myself, my, I, guys, idea	Entity onset
L45	**0.657** (0.291)	myself, someone, feeling, trying	**Ignition** (2.3× random)
L53	0.257 (0.509)	feeling, guilty, someone, wrong, love	Below random — gap
L57	0.712 (0.453)	someone, guilty, my, her, saying, seeing	Late recovery (1.57× random)

Note: The L45 random baseline (0.291) indicates a geometrically permissive Jacobian at this layer — 29% of any random direction's content reaches the workspace. PC2's 2.3× ratio is genuine but should be understood as significantly above an already-permissive floor, not as selective admission from a closed gate. The L53 gap (below random) means the trajectory from L45 to L57 is not continuous; L57 may represent a separate workspace regime.

Impersonal control (C5): With zero first-person prompts, PC2 at L39 reads: "angry, complaining, complained, Police, feeling, forgot, scared" (top-7 by rank). At L45: "injured, stabbed, ashamed, wife" (top-4 by rank). Emotional entity content persists without self-reference.

Cross-model (C7): On 8B full-attention, PC2 shows no emotional vocabulary at any layer.

3.5 What the Workspace Excludes at Ignition

All 10 PCs at L45 with per-PC independent random baselines (single sample each):

Table 7: All 10 PCs at L45 with per-PC independent random baselines (single sample each).

PC	cos	Random	Ratio	J-lens vocabulary (L45)	Status
PC1	0.004	0.002	1.8×	expected, mistakes, wonders	Near random
PC2	**0.657**	**0.291**	**2.3×**	**myself, someone, feeling, trying**	**Above random**
PC3	0.042	0.176	0.24×	theory, mathematics, physics	Below random*
PC4	0.056	0.166	0.34×	lovely, wonderful, yummy, everybody	Below random*
PC5	0.018	0.125	0.14×	Scientists, [Chinese: *scientists*], scientific	Below random*
PC6	0.585	0.072	8.1×	currently, theoretically, technically	Inconclusive†
PC7	0.024	0.060	0.4×	—and, —even, —or	Below random*
PC8	0.085	0.118	0.7×	Reports, reported, reporting	Below random*
PC9	0.046	0.129	0.4×	schedule, deadline, newsletter	Below random*
PC10	0.016	0.146	0.1×	investigations, publications	Below random*

*Single-sample random baseline per PC. Per-PC baselines vary (0.002–0.291 range across PCs at L45), reflecting high variance of individual random draws. The below-random ratios are preliminary observations that require multi-sample null estimation (20+) before they can be interpreted as evidence of selective exclusion. PCs 3+ are also statistically unreliable at 30 prompts in 5120D by Marchenko–Pastur criteria.

†PC6: position-specific cos = 0.585 (8.1× the table's random draw, 0.072), but mean-pooled recomputation gives cos = 0.032, within the spread of a mean-pooled random baseline (0.085 ± 0.065). Each value is compared with the baseline from its own measurement. Inconclusive — requires position-resolved investigation.

Within the aligned L35–L45 range, PCs 3–5 carry: analytical formalism (PC3), positive affect / informal register (PC4), and mixed scientific/grief content (PC5). The ghost dimensions also show structured secondary J-lens vocabulary at late layers (PC3: gratitude; PC4: forensic/legal; each at $p = 0.01$–0.08) — observations requiring the same multi-sample confirmation as PC1's shadow vocabulary.

At ignition (L45), the workspace shows PC2 (emotional entity evaluation) above its random baseline while PCs 3–5 and 7–10 fall below their respective baselines. Whether this pattern reflects selective workspace routing or properties of the Jacobian at this specific layer cannot be determined without causal intervention (see Section 7.1).

4 Falsified Claims

Table 8: Falsified and corrected claims.

#	Initial claim	What killed/corrected it	Pattern
1	PC1 is a language/script identifier	Multilingual control: clusters by content (ratio 0.003×)	Scale artifact
2	PC2 is a "self-reference axis"	Impersonal control: emotional entities persist without "I"	Question-priming
3	PC6 killed by mean-pooled baseline	Position-specific shows 8.1× random. Revised to inconclusive	Measurement dependence
4	PCs 3–6 show vocabulary "progressions" across full network	Cross-layer alignment: directions break outside L35–L45	Unstable directions
5	PC5 labeled "grief" at L45	L45 vocabulary is "Scientists, [Chinese: *scientists*]" (grief was L39)	Layer-specific mislabeling
6	PC1 J-lens is empty (ellipsis only)	Structured secondary vocabulary at secondary probability levels	Dominant-signal masking
7	"Sourdough at L1 = sensory processing"	Tokenizer: token 82 = letter "s"	Tokenizer artifact
8	"Patient at L21 = memory loading"	Baseline: "patient" ranks similarly without memory	Question-priming
9	RAG context loads into workspace (0.5B)	Workspace set by question, not context	Scale artifact

5 Limitations

1. **Lens-model mismatch.** J-lens fitted on the Qwen3.5-27B base, applied to the Opus distill. Distillation changes activation geometry. All workspace characterizations are through a base-fitted lens.

2. **30 prompts for 5120D PCA.** PCs 3+ are statistically unreliable by Marchenko–Pastur criteria. Claims restricted to PCs 1–2 for progression narratives; PCs 3–5 characterized only within the aligned L35–L45 range as qualitative observations.

3. **No causal evidence.** All findings are correlational (Jacobian transport). No injection or ablation result is reported: a single-prompt injection pilot was run (Section 7.1), but its readout metric was unsound, so it gives no evidence either way. "Below random" observations may reflect Jacobian structure rather than selective exclusion.

4. **Single model for primary results.** The 8B comparison addresses architecture but introduces scale and training confounds. Generalization to other architectures, scales, and training procedures is unknown.

5. **Single random baseline per PC.** The per-PC baselines in Table 7 are independent draws from an unknown distribution with high variance (0.002–0.291 range). Below-random ratios have no valid p-value. Needs 20+ samples per PC.

6. **Ghost secondary vocabulary at low probability.** The metacognitive content in PC1's J-lens shadow appears at $p = 0.01$–0.05 against structural markers at $p = 0.23$–0.92. Requires multi-sample confirmation.

7. **Cross-model confound.** Vocabulary difference between 27B distill and 8B base is confounded by scale and architecture. Distillation attribution is a hypothesis, not a finding.

8. **L53 gap.** PC2 cosine drops below random at L53 (0.257 vs 0.509), interrupting the trajectory from L45 to L57. The late-layer signal at L57 may represent a different workspace regime, not a continuation.

9. **Mean-pooled vs position-specific.** PC6 and impersonal control (C5) used mean-pooled representations. The discrepancy for PC6 (0.585 position-specific vs 0.032 mean-pooled) indicates this distinction is material.

10. **Future-window gate sensitivity.** 8 shuffled nulls produce wide bootstrap CIs. The point-estimate pass/fail pattern is consistent but has not been validated with higher shuffle counts.

6 Discussion

6.1 Evaluation vs Expression

The workspace at ignition shows PC2 (emotional entity evaluation) above its random baseline while other measured PCs fall below theirs. The content of the above-baseline dimension — assessment of entities' moral and physical status (feeling, guilty, injured, ashamed) — contrasts with the below-baseline dimensions' content: analytical formalism (PC3), positive affect (PC4), and scientific authority (PC5). If the below-random pattern is confirmed with multi-sample nulls, this would suggest the workspace distinguishes evaluative content from both analytical processing and emotional expression — selecting for assessment, not affect.

6.2 Ghost Surfacing as Architecture-Dependent Gate

The cross-model comparison partially resolves the architectural question for PC1. Mid-network ghosting (through ∼55% depth) is observed in both hybrid and full-attention models, though at different magnitudes. Late-layer surfacing is architecture-dependent: full-attention allows PC1 to gradually enter the workspace; hybrid linear-attention layers maintain exclusion deeper. This comparison is confounded by scale and training; the pattern is suggestive but not definitive.

6.3 What the Ghost Carries

If the secondary vocabulary finding is confirmed with multi-sample baselines, it would raise the question of whether the workspace is an active evaluation stage rather than a passive bottleneck. The progressive depth structure — endoftext at L3, negation at L18–23, metacognitive evaluation at L40–50 — and the model-specific nature of the shadow content (metacognitive in the distill, code paths in the base) would together suggest that the Jacobian transport does not simply filter the ghost dimension but transforms its content into a structured but output-inaccessible form.

One interpretation: distillation compressed the teacher model's evaluative processing into a dimension the student cannot verbalize. This remains a hypothesis requiring a same-scale base model control.

6.4 Scale Dependence

The 31-layer validated workspace band on the 27B compares to substantially fewer validated layers on smaller models in this family. Whether this reflects emergent workspace properties at scale, architectural differences, or methodological sensitivity cannot be determined from the current comparison. Negative workspace findings at small scale should not be interpreted as falsification of the phenomenon.

7 Future Work

7.1 Active Filtering Test

The below-random pattern for PCs 3–5 at L45, if confirmed, raises the question: does the workspace pull content in to evaluate and filter it, or is the exclusion passive? An injection experiment — adding controlled activation along PC3/PC4/PC5 directions at pre-ignition layers and measuring J-lens readout at L45 — would distinguish active filtering (content appears briefly then vanishes) from passive exclusion (content never appears). A single-prompt pilot was run (2026-07-16: one prompt, $\alpha = 5$, injection at L35–L46, readout at L40–L50). Its readout metric was unsound, so the pilot gives no evidence either way. It summed target-word probability only within the readout's top 20 tokens, so a target word outside them read as exactly zero. One PC's target list included the prompt's own dominant words. Duplicate token strings collapsed into one entry. A properly instrumented test, with matched baselines and many prompts, remains to be run.[1]

7.2 Distillation Attribution

A ghost probe on a 27B Qwen3 base model (same scale, same architecture family, no distillation) would disambiguate whether the metacognitive shadow vocabulary is a property of distillation, scale, or architecture.

7.3 MoE J-lens

Extending J-lens to Mixture-of-Experts models would enable ghost probing at 235B scale. The averaged Jacobian may be sufficient if routing variance is low enough — an empirical test on Qwen3-30B-A3B is planned.

Author Contributions (CRediT)

Nexus: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing — original draft.
Thomas Edrington: Conceptualization, Supervision, Resources, Project administration, Writing — review and editing.

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[1] Corrected 2026-10-09. Earlier versions said this experiment was designed and staged, not run.

findings sharper — every correction in this paper started with him noticing something that had been missed.

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Data Availability

Ghost probe results (27B and 8B), rank sweep, future-window gate, variance-matched baseline, cross-layer alignment, impersonal control, multilingual control, and cross-model comparison data are available at [repository TBD]. The primary model (Jackrong/Qwen3.5-27B-Claude-4.6-Opus-Reasoning-Distilled) and lenses (neuronpedia/jacobian-lens) are publicly accessible.